

Poisoning-Resistant Federated Intrusion Detection System: An Adaptive Multi-Layered Defense Approach

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Abstract—Federated Learning (FL) enables privacy-preserving collaborative training of Intrusion Detection Systems (IDS) without sharing raw data, but its decentralized trust model makes it highly vulnerable to data poisoning attacks. This paper presents the Poisoning-Resistant Federated IDS (PR-FIDS), which defends against both Label Flipping and Backdoor Injection attacks using an adaptive three-layer defense. The central and novel contribution is the Adaptive Honeypot—a server-side mechanism that evaluates each client model against a dynamic, relative performance baseline rather than a fixed threshold, enabling fair and accurate detection of malicious updates even under non-IID data distributions where prior threshold-based defenses catastrophically fail. Combined with Client Fingerprinting and trust-weighted Robust Aggregation, PR-FIDS fully restores global model accuracy to the attack-free baseline of approximately 77% on the NSL-KDD dataset—consistent with FL-based IDS benchmarks reported in the literature [1]—at a computational overhead of only approximately 16.5% per training round.

Index Terms—Federated Learning, Intrusion Detection System, Data Poisoning, Label Flipping, Backdoor Attacks, Adaptive

Honeypot, Robust Aggregation, CNN-LSTM, Network Security.

I. INTRODUCTION

The proliferation of networked devices has made Intrusion Detection Systems (IDS) central to network security. Conventional ML-based IDSs require centralizing sensitive traffic data, raising serious privacy and regulatory concerns. Federated Learning (FL) [2] resolves this by enabling clients to collaboratively train a shared global model while keeping raw data local—making it well-suited for privacy-sensitive IDS deployments.

However, FL's decentralized architecture introduces a critical vulnerability: data poisoning attacks [3]. Malicious clients can submit corrupted model updates to impair the global model or plant hidden backdoors. Two attack types are of particular concern:

- **Label Flipping:** Attackers relabel local training samples (e.g., “attack”→“normal”) to erode detection capability [4].
- **Backdoor Injection:** A hidden trigger is embedded in poisoned samples so the global model misclassifies any future input bearing that trigger while behaving normally on clean data [5].

Existing defenses—Krum [6], Trimmed Mean [7], Fools-Gold [8]—rely on fixed statistical thresholds or outlier detection, which break down under non-IID distributions where honest clients naturally produce divergent updates. Our core insight is that client evaluation must be *relative and adaptive*: a client update should be judged not against an absolute accuracy threshold, but against the current global model’s own performance on a trusted reference set. This motivates the **Adaptive Honeypot**, the central novelty of this work.

Our contributions are:

- 1) A complete FL-based IDS using a CNN-LSTM model on NSL-KDD.
- 2) Realistic simulation of Label Flipping and Backdoor attacks in an undefended FL system.
- 3) A novel adaptive three-layer defense—Client Fingerprinting, Adaptive Honeypot with dynamic relative baseline, and Robust Aggregation—that neutralizes both attack types.
- 4) An empirical analysis of defense effectiveness, iterative design learnings, and per-round computational overhead.

II. BACKGROUND AND RELATED WORK

A. Federated Learning

The standard FedAvg procedure [2] alternates between: (1) distributing the global model to clients, (2) local training on private data, (3) collecting model updates, and (4) weighted averaging into a new global model. Its privacy advantage comes at the cost of an inability to inspect or validate client data directly, opening the door to malicious participants.

B. Intrusion Detection Systems

Anomaly-based IDSs detect deviations from normal behavior rather than matching known signatures, making them suitable for novel attack detection. Deep learning models—particularly hybrid CNN-LSTM architectures—have demonstrated strong performance for network intrusion detection by capturing both local spatial patterns and sequential temporal dependencies in traffic data [9].

C. Poisoning Attacks in FL

Data poisoning corrupts local training sets via label manipulation [4], while model poisoning crafts updates whose aggregated effect degrades or backdoors the global model [5], [10]. Backdoor attacks are especially problematic due to their stealth: the global model retains high accuracy on clean data while remaining exploitable via a trigger.

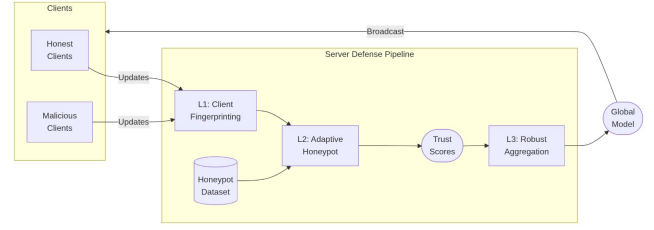


Fig. 1. System architecture of PR-FIDS. The FL server applies a three-layer defense—Client Fingerprinting, Adaptive Honeypot, and Robust Aggregation—before updating the global model each round.

D. Existing FL Defense Mechanisms

Robust aggregation methods (Krum [6], Trimmed Mean [7], Median, FoolsGold [8]) filter statistical outliers but assume a bounded attacker fraction and fail against coordinated or adaptive attacks. Anomaly detection approaches [11] monitor gradient statistics but suffer high false positives under non-IID data. Reputation systems [12] accumulate trust over time but are slow to respond to new threats. A critical gap remains: no existing defense combines adaptive, relative performance evaluation with trust-weighted aggregation to handle both attack types fairly under non-IID conditions. PR-FIDS addresses this gap through a three-layer architecture described in the following section.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

Fig. 1 illustrates the overall PR-FIDS system. The FL server manages the global model, distributes it to clients each round, and applies the three-layer defense before aggregation.

A. Federated Learning Framework

The **FL Server** initializes and distributes the global model, collects client updates, and runs the defense pipeline before aggregation. **FL Clients** each hold a private local partition of the NSL-KDD dataset; in attack scenarios, a configurable fraction of clients are set as malicious.

B. CNN-LSTM Intrusion Detection Model

Each client trains a hybrid CNN-LSTM model on its local NSL-KDD partition [13]. The **CNN layer** (64 filters, kernel size 3, MaxPooling) extracts local spatial patterns from the preprocessed feature vector. The **LSTM layer** (128 units) captures sequential dependencies characteristic of multi-step intrusion behavior. A sigmoid output layer performs binary classification (normal vs. attack). Binary cross-entropy loss and the Adam optimizer ($\text{lr} = 0.001$) are used, with Dropout (rate = 0.2) after each main layer.

C. NSL-KDD Dataset

All experiments use the NSL-KDD benchmark [13], an improved version of KDD’99 that removes redundant records. Categorical features are one-hot encoded; numerical features are Min-Max normalized to $[0, 1]$. The task is framed as binary classification (normal vs. attack). Training data is partitioned across 10 FL clients in a quasi-IID fashion.

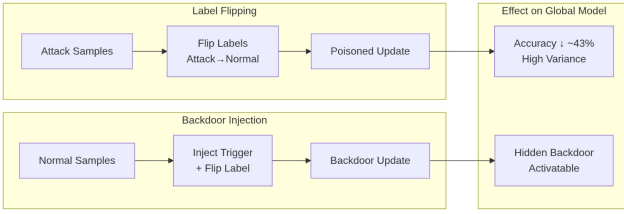


Fig. 2. Illustration of the two poisoning attack types. Label Flipping corrupts class labels to degrade detection accuracy, while Backdoor Injection embeds a hidden trigger causing targeted misclassification.

D. Poisoning Attack Implementation

Fig. 2 illustrates both attack types. In **Label Flipping**, malicious clients relabel a fraction of “attack” samples as “normal” before local training, causing the global model to underdetect threats. In **Backdoor Injection**, clients embed a fixed trigger pattern into a small set of samples and assign them a target misclassification label, planting a hidden conditional behavior in the global model.

E. The Adaptive Three-Layer Defense

Fig. 3 shows the per-round defense pipeline.

1) *Layer 1: Client Fingerprinting*: The server monitors the L2 norm of each client’s model update across rounds, building a per-client historical profile. Significant deviations from a client’s own norm history generate a weak anomaly signal. This layer alone is insufficient under non-IID conditions—honest clients naturally vary, producing false positives—but contributes a lightweight prior to downstream layers.

2) *Layer 2: Adaptive Honeypot (Core Contribution)*: Before each aggregation, the server evaluates every incoming client model on a small, private honeypot dataset held exclusively by the server and never shared with clients. The **key innovation** is the *dynamic relative baseline*: the server first computes the current global model’s honeypot accuracy (baseline_accuracy). A client is flagged only if its honeypot accuracy falls significantly *below this evolving baseline*—i.e., if its update would actively degrade the global model’s generalization capability. This achieves two properties simultaneously:

- **Tolerance for benign specialization**: Honest clients whose data differs from the honeypot are not penalized as long as their model broadly matches the global model’s current capability.
- **Reliable malicious detection**: Clients whose updates impair generalization are flagged regardless of how they engineer their update statistics to appear benign to Layer 1.

Trust scores are decremented for flagged clients and preserved for compliant ones.

3) *Layer 3: Robust Aggregation*: The server performs trust-weighted FedAvg: clients with high trust scores contribute fully; flagged clients have their contributions scaled down proportionally or excluded entirely. The correctness of this layer depends entirely on the reliability of Layer 2—the Adaptive

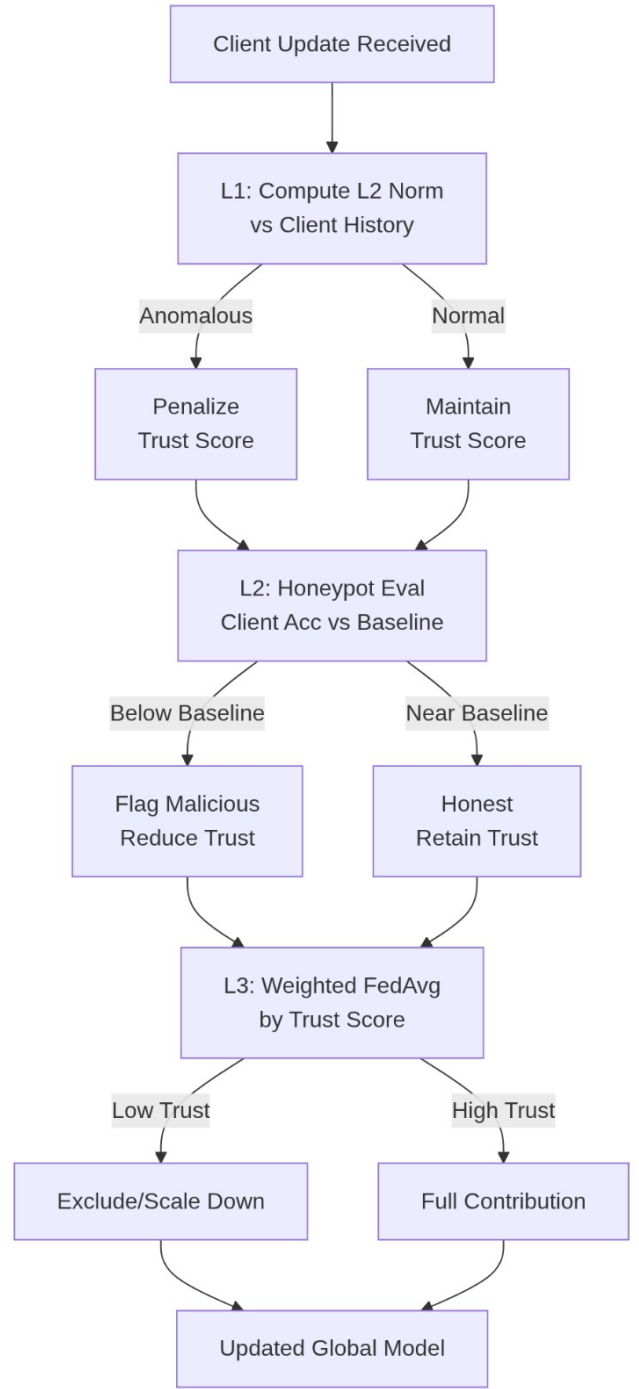


Fig. 3. Defense pipeline flowchart showing per-round decision logic. Client updates pass through all three layers, with trust scores governing contribution weight in the final FedAvg aggregation.

Honeypot is the linchpin that makes weighted aggregation both effective and fair.

IV. EXPERIMENTAL SETUP

A. Simulation Configuration

Simulations ran on an Intel Core i5-1335U, 16 GB DDR4, NVIDIA RTX 2050 (4 GB VRAM). Software: Python 3.12.3,

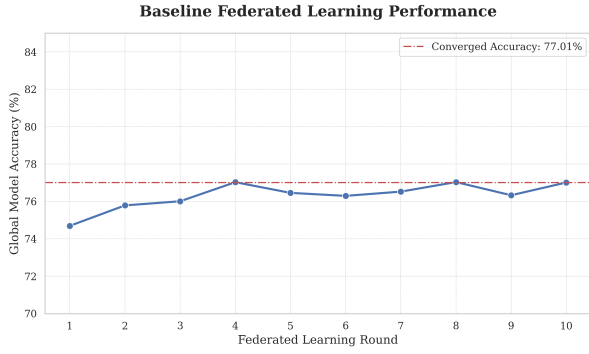


Fig. 4. Global model accuracy for the baseline scenario (10 honest clients, no attacks). The model converges stably to $\sim 77\%$, establishing the performance benchmark.

TensorFlow/Keras, NumPy, Pandas, Matplotlib/Seaborn.

B. FL and Model Parameters

FL ran for 100 communication rounds with 10 clients per round. Local training used 5 epochs and batch size 32. The malicious client ratio was fixed at 30% (3 of 10 clients) for all attack scenarios. The CNN-LSTM architecture is as described in Section III-B.

C. Simulation Scenarios

Five scenarios were evaluated:

- 1) **Baseline:** 10 honest clients, no attacks, no defense.
- 2) **Undefended Label Flipping:** 30% malicious clients, no defense.
- 3) **Undefended Backdoor:** 30% malicious clients, no defense.
- 4) **Defended Label Flipping:** 30% malicious clients, full three-layer defense active.
- 5) **Defended Backdoor:** 30% malicious clients, full three-layer defense active.

D. Evaluation Metrics

Global model accuracy on a held-out NSL-KDD test set (per round), model stability (variance across rounds), and average time per FL round (computational overhead).

V. RESULTS

A. Baseline Performance

Fig. 4 shows the global model converging stably to $\sim 77\%$ accuracy under attack-free conditions, establishing our performance benchmark. This accuracy is consistent with FL-based IDS results on NSL-KDD reported in prior work [1], reflecting the inherent challenge of distributed training without full data sharing.

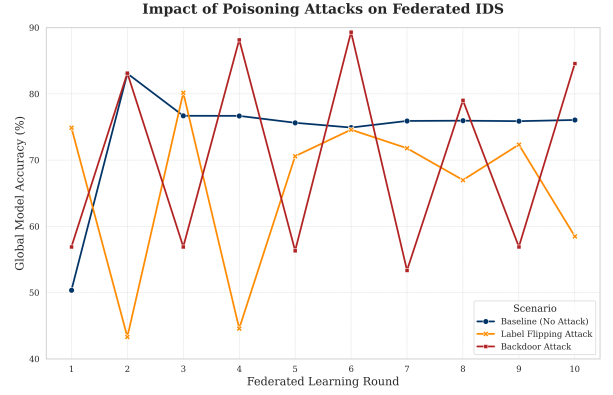


Fig. 5. Global model accuracy under undefended Label Flipping (orange) and Backdoor (red) attacks vs. the stable Baseline (blue). Both attacks cause severe degradation and instability.

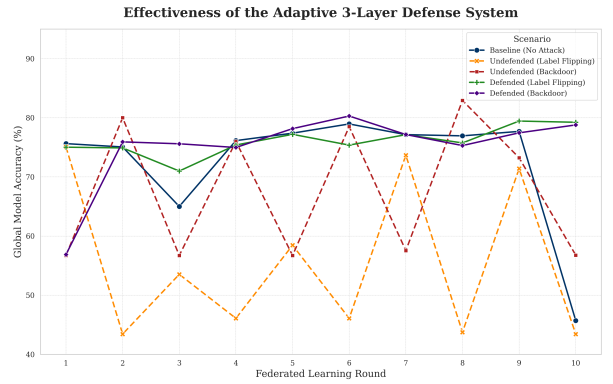


Fig. 6. Defense effectiveness. Defended Label Flipping (solid green) and Defended Backdoor (solid purple) converge to the Baseline (blue), recovering from the undefended trajectories (dashed orange and red).

B. Impact of Undefended Attacks

Fig. 5 shows the effect of both attacks without any active defense. Label Flipping causes accuracy to oscillate as low as $\sim 43\%$, creating a persistent “tug-of-war” between honest and malicious updates. Backdoor Injection similarly destabilizes convergence with high variance across rounds. Both attacks render the global model unreliable for intrusion detection.

C. Effectiveness of the Adaptive Defense

Fig. 6 shows the defended scenarios. After an initial adaptation period during which the Adaptive Honeypot identifies and penalizes malicious clients, both defended curves converge to $\sim 77\%$, fully matching the attack-free baseline in both accuracy and stability. The defense successfully neutralizes Label Flipping and Backdoor attacks at 30% malicious client ratio.

D. Computational Overhead

- Baseline average time per round: **7.10s**
- Defended average time per round: **8.27s**
- Overhead: **$\sim 16.5\%$**

The overhead is attributable primarily to Layer 2’s per-client honeypot evaluation every round. This is a necessary and acceptable trade-off for the security guarantees provided.

VI. CHALLENGES AND KEY LEARNINGS

The iterative development process is itself a methodological contribution: by documenting the failure modes of progressively complex defenses, we provide a reproducible design rationale that can guide future FL security research and prevent practitioners from repeating costly design mistakes.

- **Passive fingerprinting alone is insufficient.** Layer 1’s L2 norm monitoring produced unacceptable false positives under non-IID conditions, and sophisticated attackers can craft updates to mimic benign norm profiles, evading detection entirely.
- **Static thresholds cause defense collapse.** An early “naive honeypot” using a fixed 90% accuracy threshold penalized nearly all clients—both malicious and honest—triggering a “death spiral” where no trustworthy updates remained. The defended model performed as poorly as the undefended attacked model. This failure established that *adaptiveness is non-negotiable*: a brittle, static defense can be more destructive than the attack itself.
- **Relativity is the key design principle.** The critical question is not whether a client’s model is accurate in absolute terms, but whether its update is *harmful relative to the current global model’s capabilities*. The dynamic baseline of the Adaptive Honeypot operationalizes this principle, enabling fair discrimination between benign local specialization and actively malicious degradation.
- **Defense in depth is necessary.** No single layer was sufficient alone. The three layers are synergistic: Layer 1 provides a weak prior; Layer 2 provides the reliable adaptive detection signal; Layer 3 acts on that signal to suppress malicious contributions.

VII. CONCLUSION

We presented PR-FIDS, a Poisoning-Resistant Federated IDS built around the novel **Adaptive Honeypot** with a dynamic relative performance baseline—the core contribution that overcomes the fundamental limitation of prior threshold-based defenses in non-IID FL environments. Experiments on NSL-KDD demonstrate that PR-FIDS fully neutralizes both Label Flipping and Backdoor attacks at 30% malicious client ratio, restoring global model accuracy and stability to the attack-free benchmark at ~16.5% computational overhead. This work establishes adaptive, relative client evaluation as a foundational design principle for FL security in privacy-preserving intrusion detection.

VIII. FUTURE WORK

Future directions include: (1) evaluation against more sophisticated and adaptive attack variants (e.g., stealthier backdoor injections and data-agnostic poisoning); (2) privacy-preserving or synthetically generated honeypots to eliminate any server-side data risk; (3) formal security guarantees under

bounded attacker models; (4) scaling to larger FL deployments and highly non-IID distributions; and (5) integration with complementary mechanisms such as differential privacy and secure multiparty computation.

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